

Exercise 1: Email Search Optimization

A) Identify a way to optimize the search performance for customer emails

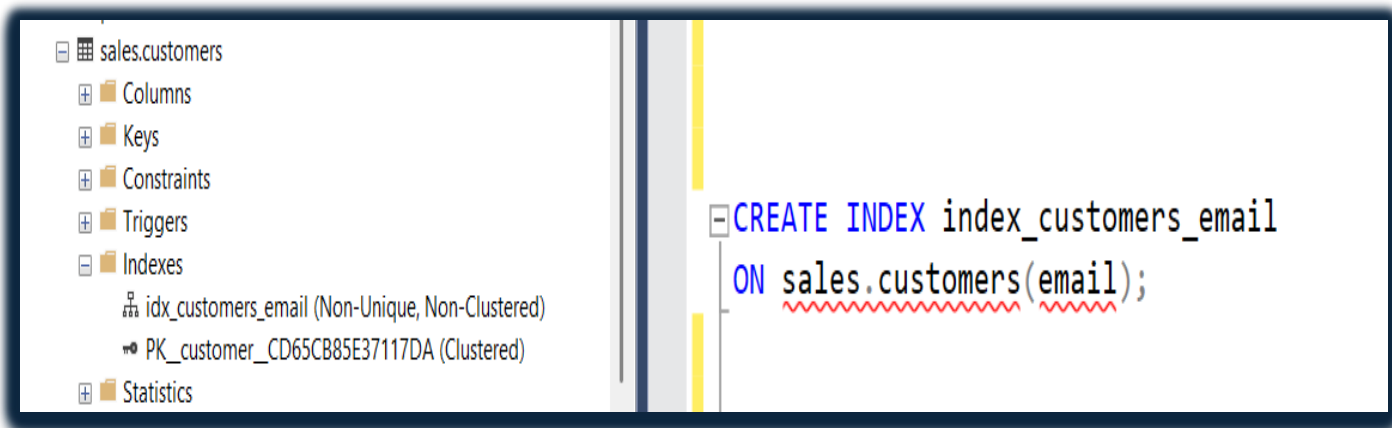
To optimize search performance when looking up customers by their email addresses, we can create an **index** on the email column in the customers table.

An index improves query performance by allowing the database engine to locate the required rows faster instead of scanning the entire table.

B) Implement the optimization on the customers table

```
CREATE INDEX index_customers_email
```

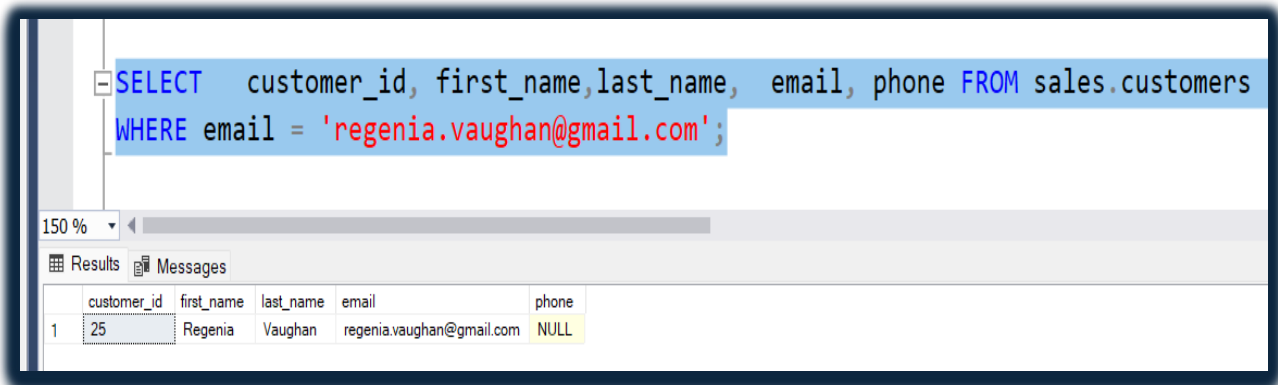
```
ON sales.customers(email);
```



C) Retrieve customer details using an email filter

```
SELECT customer_id, first_name, last_name, email, phone FROM sales.customers
```

```
WHERE email = 'regenia.vaughan@gmail.com';
```



Exercise 2: Order Status Index

A) Create an Index to improve the performance of these queries

```
CREATE INDEX index_orders_customer_status  
ON sales.orders(customer_id, order_status);
```



B) Retrieve order details for a customer's shipped orders

The following query retrieves the order information for a specific customer where the order status is 'Shipped'== 4

```
SELECT order_id, order_date, required_date, shipped_date, store_id  
FROM sales.orders  
WHERE customer_id = 2  
AND order_status = 4;
```

The screenshot shows a SQL query window with the following text:

```
SELECT
order_id, order_date, required_date,
shipped_date, store_id
FROM sales.orders
WHERE customer_id = 2
AND order_status = 4;
```

Below the query window, the Results pane is visible, showing a single row of data:

	order_id	order_date	required_date	shipped_date	store_id
1	1084	2017-08-21	2017-08-24	2017-08-23	1

Exercise 3: Inventory View

A) Create a View to simplify future inventory lookups

create view inventory_view as

SELECT s.store_name, p.product_name, st.quantity

FROM production.stocks st

JOIN sales.stores s ON st.store_id = s.store_id

JOIN production.products p ON st.product_id = p.product_id;

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Views' folder is expanded, showing the 'dbo.inventory_view' view. The 'Columns' folder is expanded, showing the following columns:

- store_name (varchar(255), not null)
- product_name (varchar(255), not null)
- quantity (int, null)

On the right, the SQL query window shows the following text:

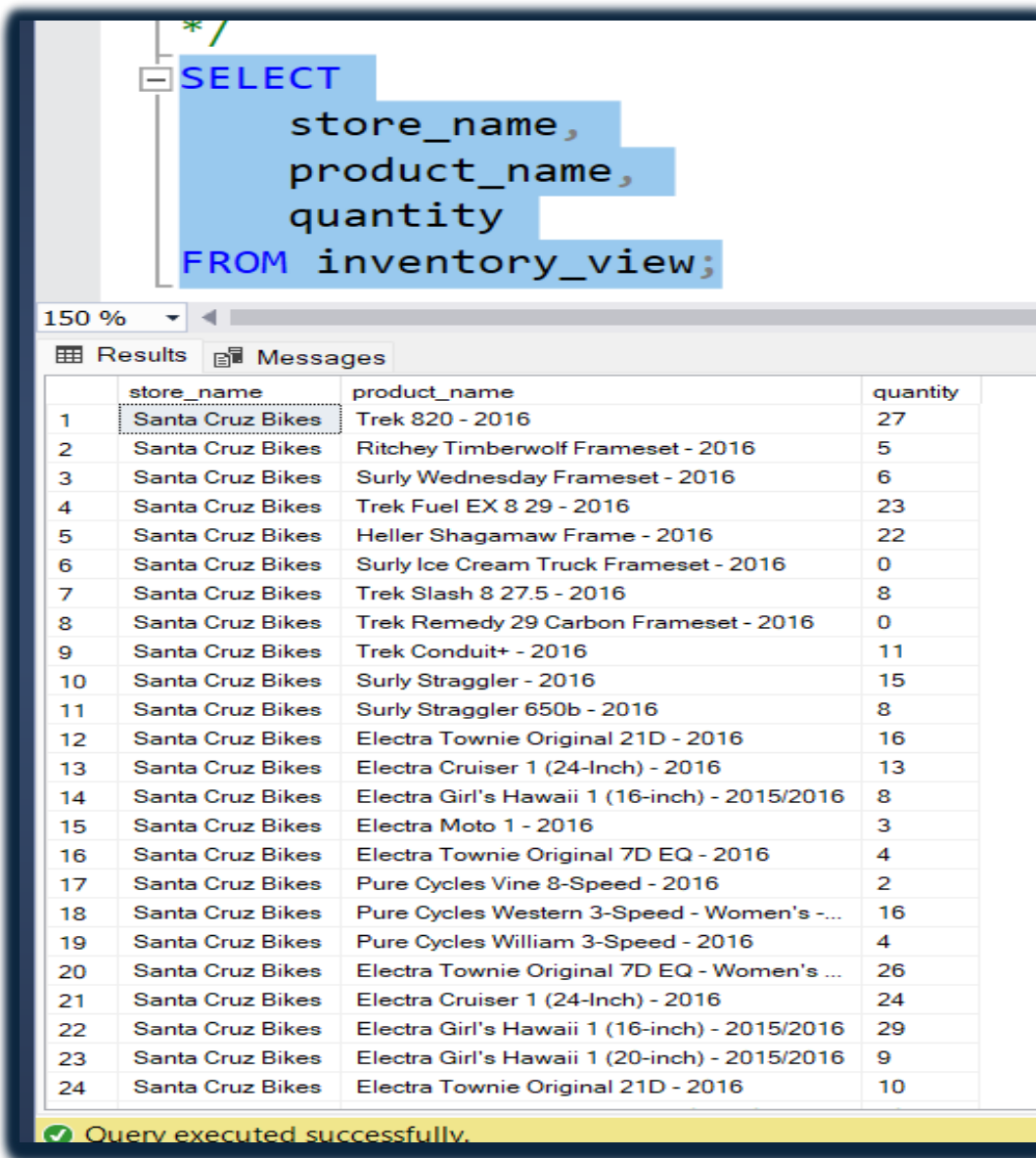
```
create view inventory_view as
SELECT
s.store_name,
p.product_name,
st.quantity
FROM production.stocks st
JOIN sales.stores s ON st.store_id = s.store_id
JOIN production.products p ON st.product_id = p.product_id;
```

B) Retrieve and display the inventory data for all stores

After creating the view, we can easily retrieve the inventory report using the following query:

```
SELECT store_name, product_name, quantity
```

```
FROM inventory_view;
```



The screenshot shows a SQL query execution interface. At the top, the query is displayed in a text editor with syntax highlighting: `SELECT store_name, product_name, quantity FROM inventory_view;`. Below the query editor, there is a tabbed interface with 'Results' and 'Messages' tabs. The 'Results' tab is active, showing a table with 24 rows and 3 columns: `store_name`, `product_name`, and `quantity`. The first row is highlighted. At the bottom, a yellow status bar indicates 'Query executed successfully.'

	store_name	product_name	quantity
1	Santa Cruz Bikes	Trek 820 - 2016	27
2	Santa Cruz Bikes	Ritchey Timberwolf Frameset - 2016	5
3	Santa Cruz Bikes	Surly Wednesday Frameset - 2016	6
4	Santa Cruz Bikes	Trek Fuel EX 8 29 - 2016	23
5	Santa Cruz Bikes	Heller Shagamaw Frame - 2016	22
6	Santa Cruz Bikes	Surly Ice Cream Truck Frameset - 2016	0
7	Santa Cruz Bikes	Trek Slash 8 27.5 - 2016	8
8	Santa Cruz Bikes	Trek Remedy 29 Carbon Frameset - 2016	0
9	Santa Cruz Bikes	Trek Conduit+ - 2016	11
10	Santa Cruz Bikes	Surly Straggler - 2016	15
11	Santa Cruz Bikes	Surly Straggler 650b - 2016	8
12	Santa Cruz Bikes	Electra Townie Original 21D - 2016	16
13	Santa Cruz Bikes	Electra Cruiser 1 (24-Inch) - 2016	13
14	Santa Cruz Bikes	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	8
15	Santa Cruz Bikes	Electra Moto 1 - 2016	3
16	Santa Cruz Bikes	Electra Townie Original 7D EQ - 2016	4
17	Santa Cruz Bikes	Pure Cycles Vine 8-Speed - 2016	2
18	Santa Cruz Bikes	Pure Cycles Western 3-Speed - Women's -...	16
19	Santa Cruz Bikes	Pure Cycles William 3-Speed - 2016	4
20	Santa Cruz Bikes	Electra Townie Original 7D EQ - Women's ...	26
21	Santa Cruz Bikes	Electra Cruiser 1 (24-Inch) - 2016	24
22	Santa Cruz Bikes	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	29
23	Santa Cruz Bikes	Electra Girl's Hawaii 1 (20-inch) - 2015/2016	9
24	Santa Cruz Bikes	Electra Townie Original 21D - 2016	10